

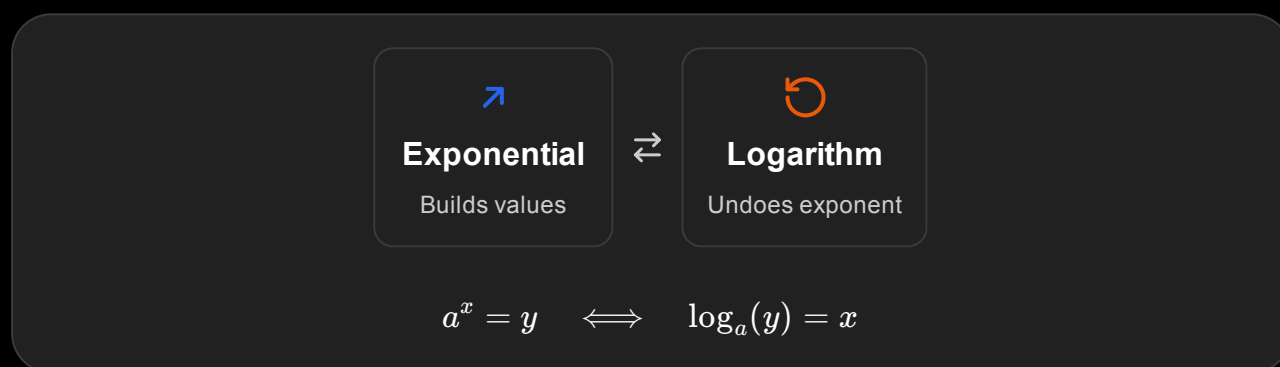


You're mixing **four related but different ideas**:

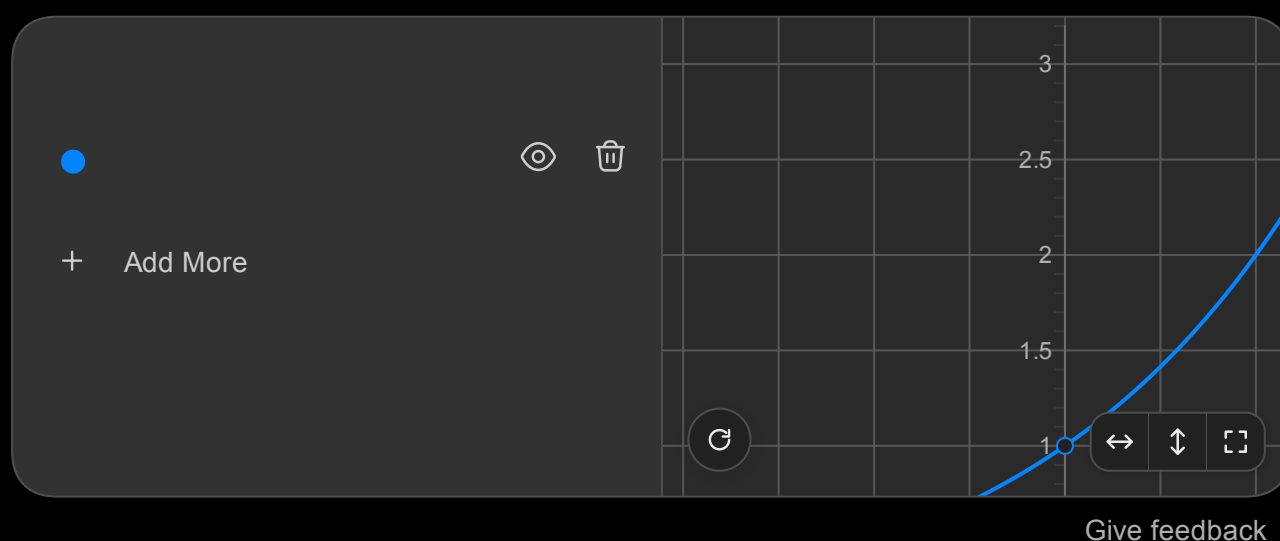
- **Exponential function** → “raising a base to a power”
- **Logarithm** → “undoing an exponent”
- **Natural logarithm (ln)** → logarithm with base **e**
- **e** → a special mathematical constant ≈ 2.71828

Think of them as pairs.

The big picture



1. Exponential: base raised to a power



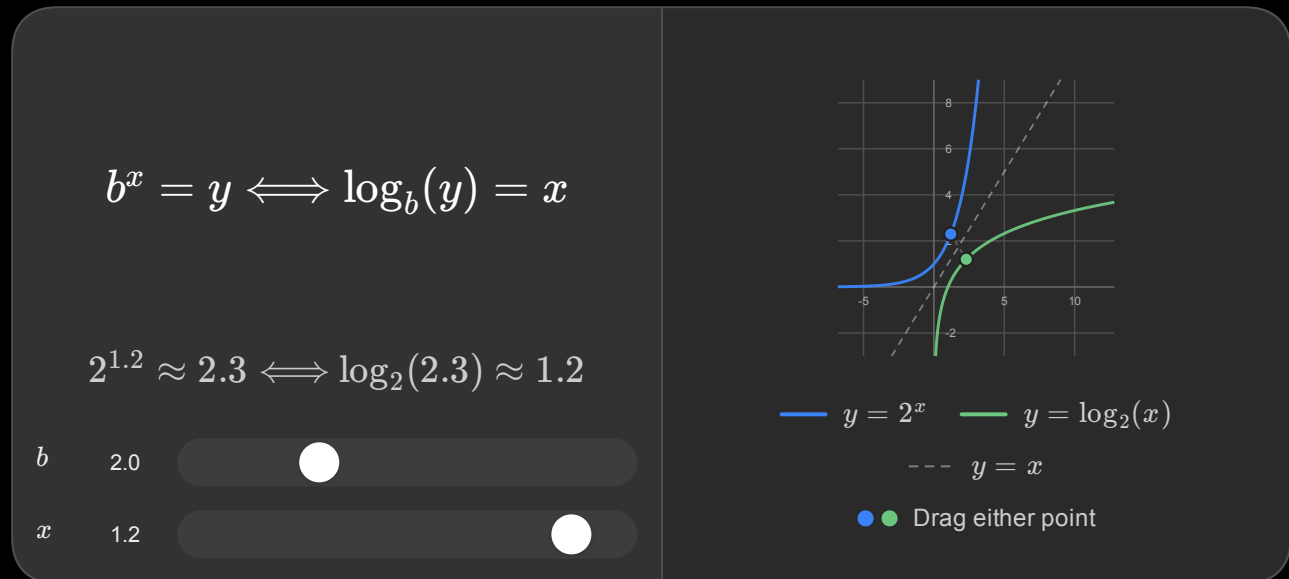
This means:

- $2^3 = 8$
- $10^2 = 100$

- $e^1 = e$

The **base** can be 2, 10, e, or any positive number except 1.

2. Logarithm: asks “what power gave this number?”



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Examples:

- $\log_2(8) = 3$ because $2^3 = 8$
- $\log_{10}(100) = 2$ because $10^2 = 100$

So logarithms are the **inverse** of exponentials.

3. Natural logarithm (ln)

This is just a logarithm with base **e**.

$$\ln(x) = \log_e(x)$$

Examples:

- $\ln(e) = 1$
- $\ln(e^3) = 3$
- $e^{\ln(5)} = 5$

4. What is e?

A special irrational constant:

$$e \approx 2.718281828 \dots$$

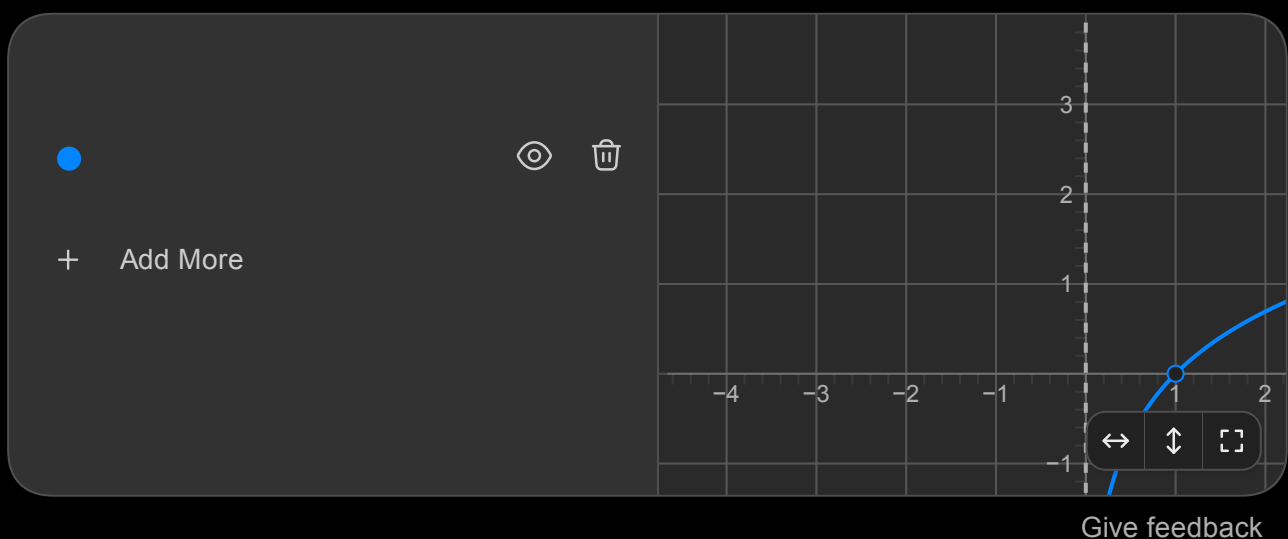
It appears naturally in:

- continuous growth
- compound interest
- calculus
- differential equations
- machine learning and statistics

The most important exponential is:



And its inverse is:



Compare them directly

Expression	Meaning
2^x	Exponential with base 2
10^x	Exponential with base 10
e^x	Natural exponential
$\log_2(x)$	Log base 2
$\log_{10}(x)$	Common log (often written $\log(x)$)
$\ln(x)$	Natural log = log base e

One example connecting all of them

Suppose:

$$e^x = 20$$

To solve for x, apply $\ln$ to both sides:

$$\ln(e^x) = \ln(20)$$

Since $\ln$ undoes e^x :

$$x = \ln(20) \approx 2.996$$

Similarly:

$$10^x = 20$$

Then:

$$x = \log_{10}(20) \approx 1.301$$

Notice the answer is different because the **base is different**.

Why does calculus love $\ln$ and e ?

The magical property:

$$\frac{d}{dx} e^x = e^x$$

and

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

No other base is this clean. For base a :

$$\frac{d}{dx} a^x = a^x \ln(a)$$

That extra $\ln(a)$ is why mathematicians prefer e .

In machine learning / regression

You'll mostly see:

- **Exponential growth/decay:** $y = e^{kx}$
- **Log transform:** $\ln(y)$
- **Softmax:** uses e^x
- **Log-likelihood:** uses $\ln$

Memory trick

Think of it as a ladder

e

special constant

↓

e^x

natural exponential



$$\ln(x)$$

undoes e^x

And separately:

$$a^x$$

any base a



$$\log_a(x)$$

undoes a^x

The 10-second exam answer

- **e**: a special constant ≈ 2.718 .
- **Exponential**: a^x (growth/decay).
- **Log base n**: $\log_n(x)$ asks “to what power must n be raised to get x ?”
- **Natural log**: $\ln(x) = \log_e(x)$.
- **Inverse relationship**:

$$a^x = y \iff \log_a(y) = x$$

and especially

$$e^x = y \iff \ln(y) = x$$